

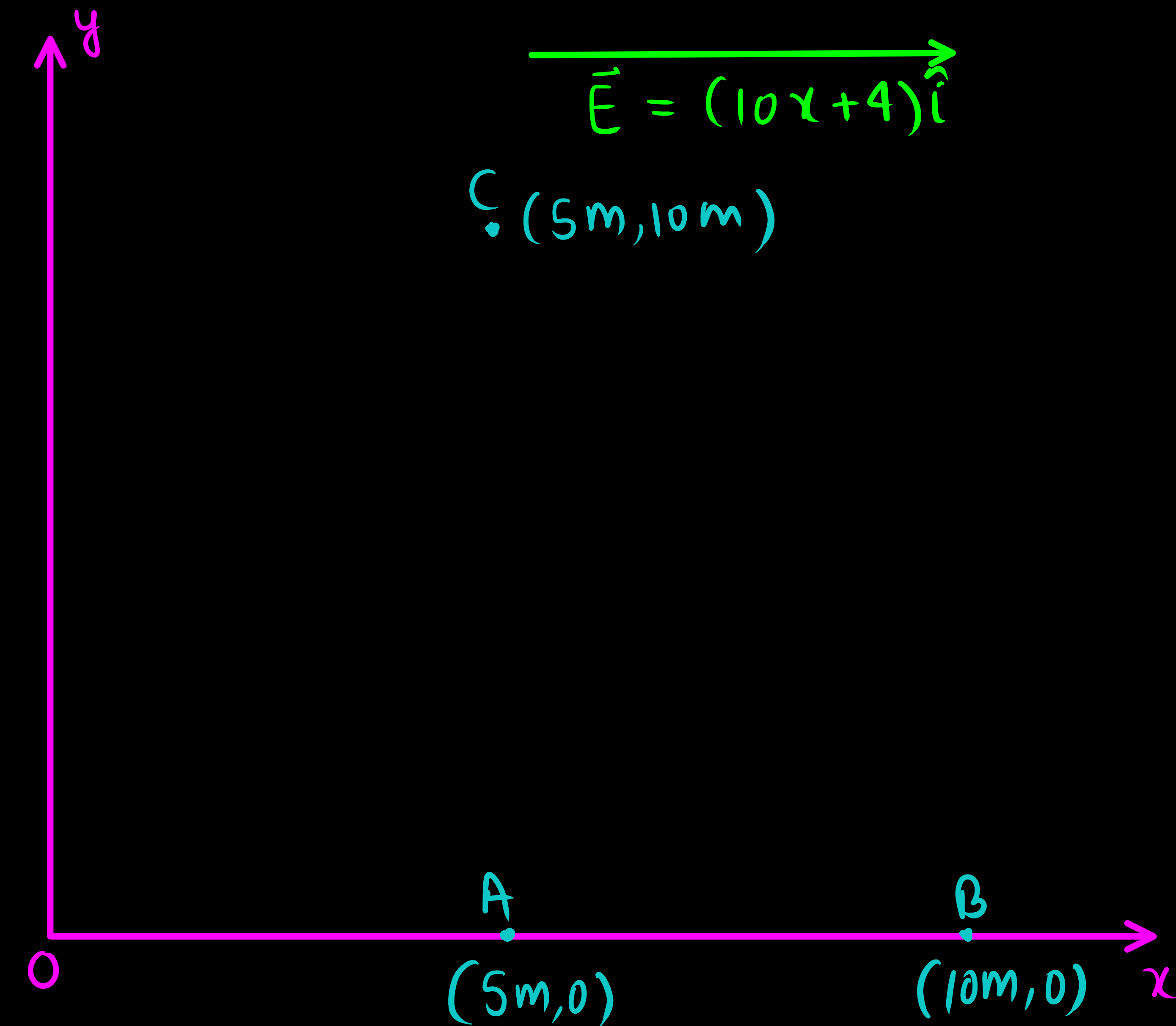
23. The electric field in a region is given by

$$\vec{E} = (10x + 4) \hat{i}$$

where x is in m and E is in N/C. Calculate the amount of work done in taking a unit charge from

- (i) (5 m, 0) to (10 m, 0)
- (ii) (5 m, 0) to (5 m, 10 m)

3



- 31.** (a) (i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.
- (ii) A charge of $6\ \mu\text{C}$ is given to a hollow metallic sphere of radius $0.2\ \text{m}$. Find the potential at (i) the surface and (ii) the centre of the sphere.

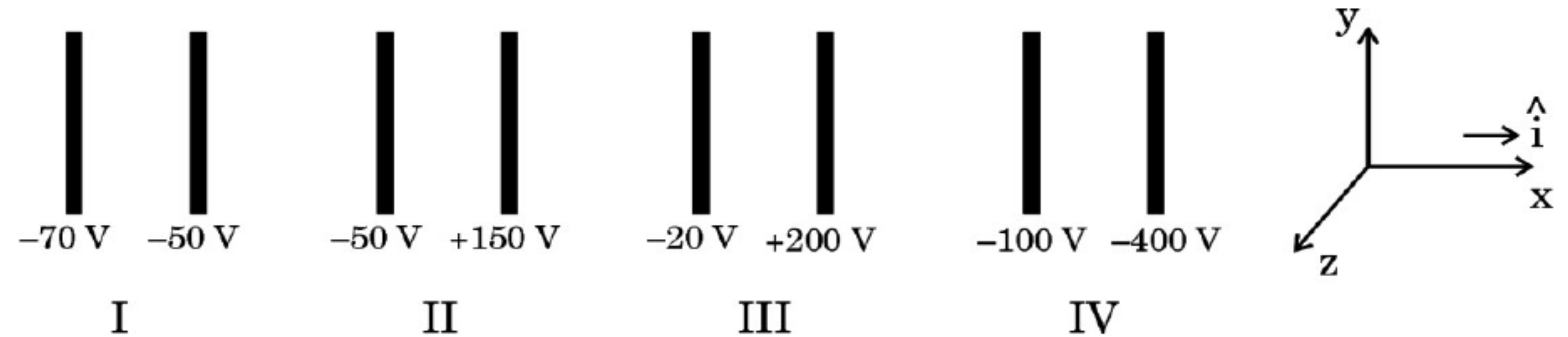
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OR

- (b) (i) A charge $+Q$ is placed on a thin conducting spherical shell of radius R . Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.
- (ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet.

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29. The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2.0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates.



- (i) For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$? 1
- (A) I (B) II
- (C) III (D) IV
- (ii) An electron is released midway between the plates of pair IV. It will : 1
- (A) move along $\hat{i}$ at constant speed
- (B) move along $-\hat{i}$ at constant speed
- (C) accelerate along $\hat{i}$
- (D) accelerate along $-\hat{i}$

- (iii) Let V_0 be the potential at the left plate of any set, taken to be at $x = 0$ m. Then potential V at any point ($0 \leq x \leq 2$ cm) between the plates of that set can be expressed as :

1

- (A) $V = V_0 + \alpha x$ (B) $V = V_0 + \alpha x^2$
(C) $V = V_0 + \alpha x^{1/2}$ (D) $V = V_0 + \alpha x^{3/2}$

where α is a constant, positive or negative.

- (iv) (a) Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then :

1

- (A) $E_1 > E_2 > E_3 > E_4$ (B) $E_3 > E_4 > E_1 > E_2$
(C) $E_4 > E_3 > E_2 > E_1$ (D) $E_2 > E_3 > E_4 > E_1$

OR

- (b) An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about :

(Take $(e/m) = 1.76 \times 10^{11}$ C/kg)

1

- (A) 1.3×10^5 m/s (B) 2.6×10^6 m/s
(C) 6.5×10^5 m/s (D) 5.2×10^7 m/s

1. The plates P_1 and P_2 of a $2\ \mu\text{F}$ capacitor are at potentials $25\ \text{V}$ and $-25\ \text{V}$ respectively. The charge on plate P_1 will be :

(A) $0.02\ \text{mC}$

(B) $0.1\ \text{mC}$

(C) $0.1\ \mu\text{C}$

(D) $1\ \mu\text{C}$

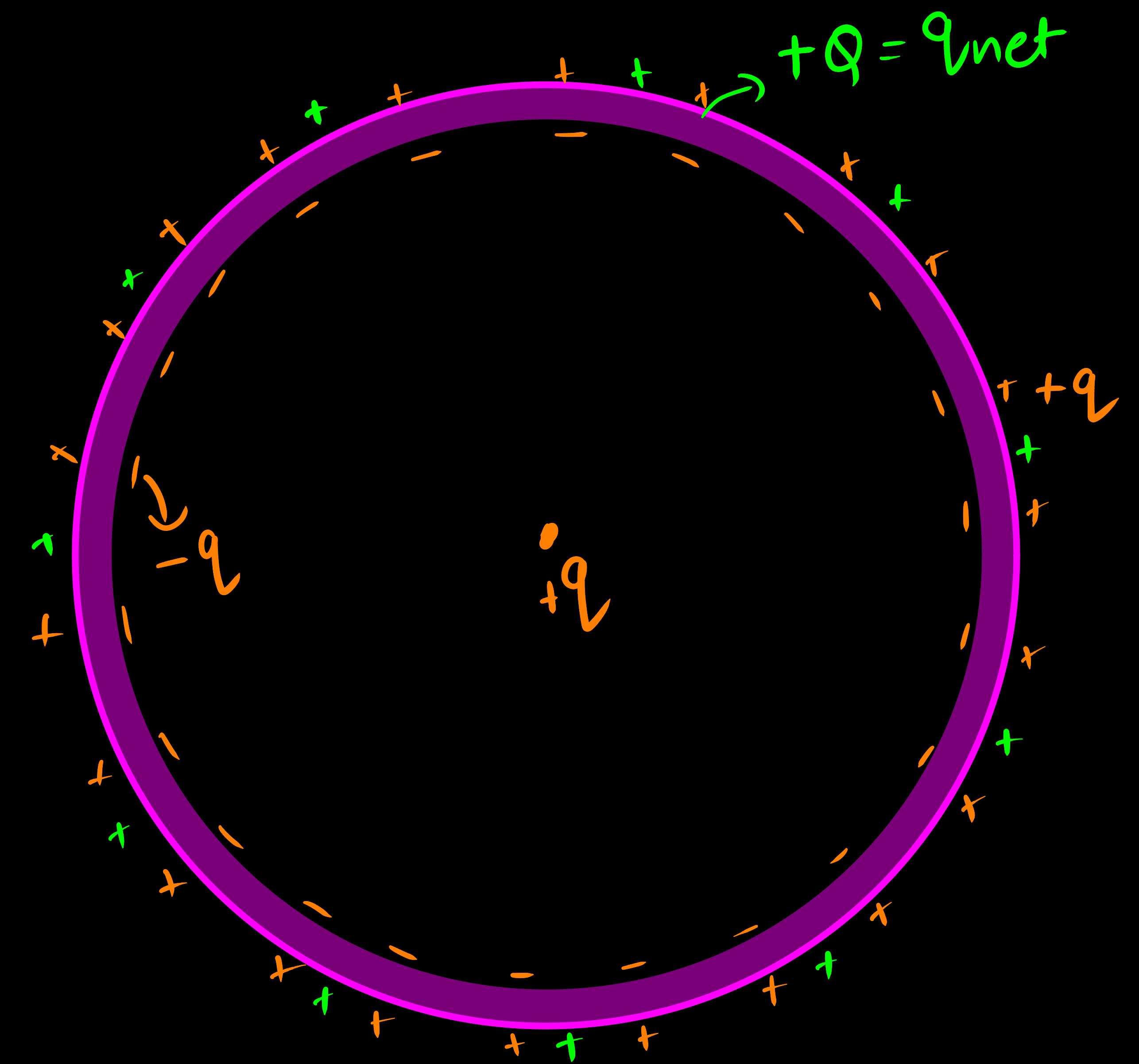
1. An isolated conductor, with a cavity, has a net charge $+Q$. A point charge $+q$ is inside the cavity. The charges on the cavity wall and the outer surface are respectively :

(A) 0 and Q

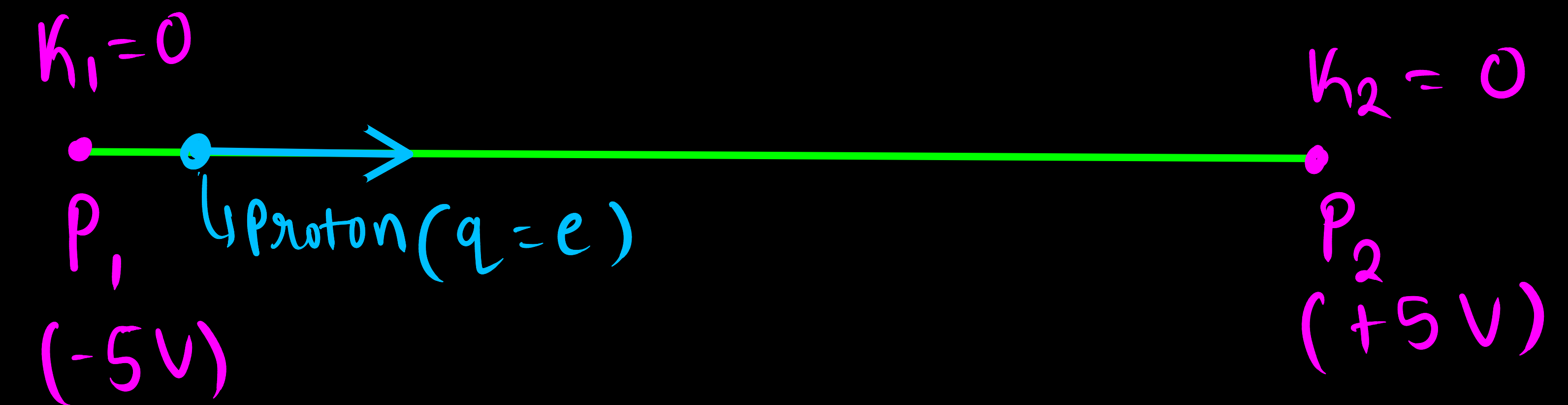
(B) $-q$ and $Q - q$

(C) $-q$ and $Q + q$

(D) 0 and $Q - q$



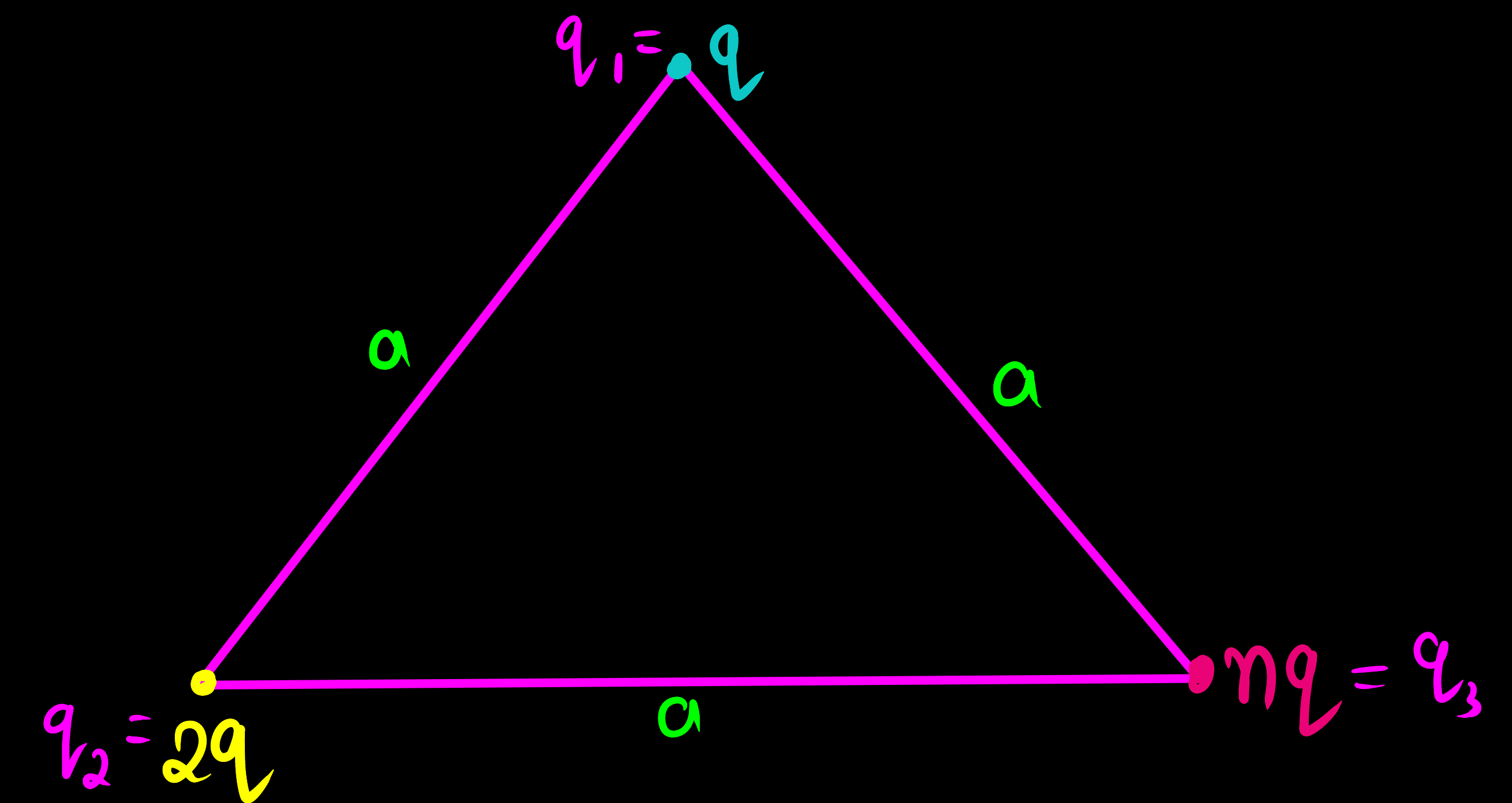
2. A proton is taken from point P_1 to point P_2 , both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5\text{ V}$ respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :
- (A) $-1.6 \times 10^{-18}\text{ J}$ (B) $1.6 \times 10^{-18}\text{ J}$
(C) Zero (D) $0.8 \times 10^{-18}\text{ J}$



1. Consider a group of charges $q_1, q_2, q_3 \dots$ such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :
- (A) Plane (B) Spherical surface
(C) Paraboloidal surface (D) Ellipsoidal surface

- 32.** (a) (i) Obtain an expression for the electric potential due to a small dipole of dipole moment $\vec{p}$, at a point $\vec{r}$ from its centre, for much larger distances compared to the size of the dipole.
- (ii) Three point charges q , $2q$ and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n .

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1. The capacitance of a parallel plate capacitor having a medium of dielectric constant $K = 4$ in between the plates is C . If this medium is removed, then the capacitance of the capacitor becomes :

(A) $4C$

(B) C

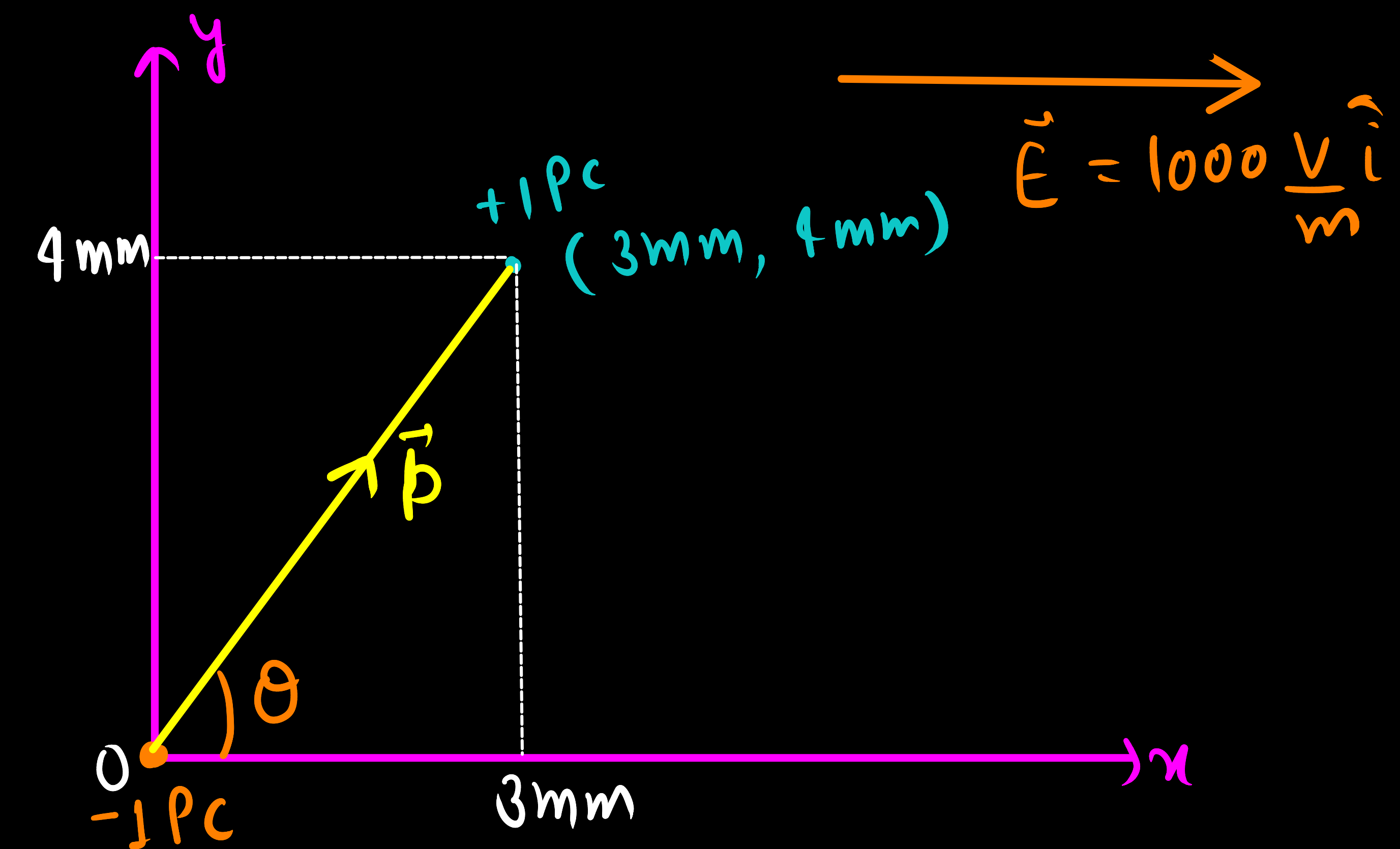
(C) $\frac{C}{4}$

(D) $2C$

- 32.** (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum ?
- (ii) An electric dipole consists of point charges -1.0 pC and $+1.0 \text{ pC}$ located at $(0, 0)$ and $(3 \text{ mm}, 4 \text{ mm})$ respectively in $x - y$ plane. An electric field $\vec{E} = \left(\frac{1000 \text{ V}}{\text{m}}\right) \hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.

5

OR



- (b) (i) An electric dipole (dipole moment $\vec{p} = p \hat{i}$), consisting of charges $-q$ and q , separated by distance $2a$, is placed along the x -axis, with its centre at the origin. Show that the potential V , due to this dipole, at a point x , ($x \gg a$) is equal to $\frac{1}{4\pi\epsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$.
- (ii) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) \text{C/m}^2$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1 .

22. Three point charges Q_1 , Q_2 and Q_3 are located in $x - y$ plane at points $(-d, 0)$, $(0, 0)$ and $(d, 0)$ respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ?

2. Ten capacitors, each of capacitance $1\ \mu\text{F}$, are connected in parallel to a source of $100\ \text{V}$. The total energy stored in the system is equal to :

(A) $10^{-2}\ \text{J}$

(B) $10^{-3}\ \text{J}$

(C) $0.5 \times 10^{-3}\ \text{J}$

(D) $5.0 \times 10^{-2}\ \text{J}$

31. (a) (i) A capacitor of capacitance C is charged to a potential difference V by a battery. The battery is disconnected and the separation between the plates is halved. How will the following quantities be affected ?

(1) Capacitance of the capacitor

(2) Electric field between the plates

(3) Energy stored in the capacitor

Justify your answer in each case.

(ii) A capacitor of $20\ \mu\text{F}$ is charged to $30\ \text{V}$. It is then connected to an uncharged $30\ \mu\text{F}$ capacitor. Calculate the charges on each capacitor in equilibrium.

OR

- (b) (i) Obtain an expression for electric potential at a distance r from a point charge Q .
- (ii) Two point charges of $10\ \mu\text{C}$ and $-5\ \mu\text{C}$ are placed at $(-3\ \text{cm}, 0)$ and $(6\ \text{cm}, 0)$ in an external electric field $E = \frac{A}{r^2}$, where $A = 1.8 \times 10^5\ \text{N C}^{-1}\ \text{m}^2$. Calculate the electrostatic energy of the system.

(iv) (b) A electric dipole consists of two point charges $+2\text{ pC}$ and -2 pC , separated by a distance 1 mm . It is kept in a uniform electric field of $2 \times 10^5\text{ NC}^{-1}$. The amount of work done in rotating it from its position of stable to unstable equilibrium will be : *1*

(A) $2 \times 10^{-10}\text{ J}$

(B) $4 \times 10^{-10}\text{ J}$

(C) $8 \times 10^{-10}\text{ J}$

(D) $1.6 \times 10^{-11}\text{ J}$

(iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be : *1*

(A) p

(B) $2p$

(C) $p\sqrt{2}$

(D) $\frac{p}{\sqrt{2}}$

8. Two particles A and B of the same mass but having charges q and $4q$ respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :

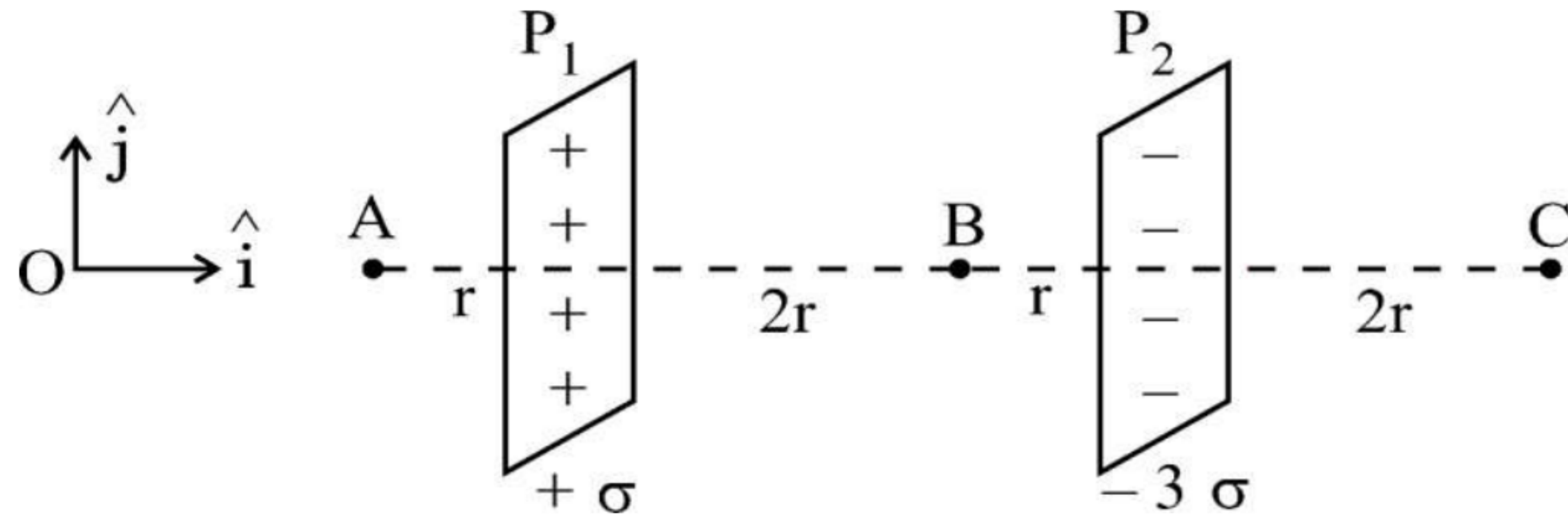
(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 2

(D) 4

- (b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C.



23. Two point charges of $10\ \mu\text{C}$ and $20\ \mu\text{C}$ are located at points $(-4\ \text{cm}, 0, 0)$ and $(5\ \text{cm}, 0, 0)$ respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6\ \text{NC}^{-1}\ \text{m}^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. 3

1. Three point charges $+q$, $-4q$ and $+2q$ are kept at the vertices of an equilateral triangle of side 'a'. The electrostatic potential energy of this system of charges is :

(A) $-\frac{1}{4\pi\epsilon_0}\left(\frac{q^2}{a^2}\right)$

(B) $+\frac{1}{4\pi\epsilon_0}\left(\frac{8q^2}{a^2}\right)$

(C) $+\frac{1}{4\pi\epsilon_0}\left(\frac{6q^2}{a}\right)$

(D) $-\frac{1}{4\pi\epsilon_0}\left(\frac{10q^2}{a}\right)$

25. Explain the polarization of a dielectric medium filled inside a charged capacitor. How does the capacity of a capacitor increase due to filling ?
What is dielectric constant of a dielectric ? 3

29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.

(i) The shape of equipotential surface due to an isolated point charge is :

- (A) ellipsoidal, with charge at its one foci
- (B) plane surface not passing through point charge
- (C) spherical with charge at its centre
- (D) cylindrical with charge at its axis

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are :

- (A) planes parallel to the XY plane
- (B) concentric spherical surfaces
- (C) planes parallel to the YZ plane
- (D) planes parallel to the XZ plane

(iii) Which of the following statements is *not* true for equipotential surfaces ? 1

- (A) Potentials at two equipotential surfaces are different.
- (B) No work is done in moving a charge on an equipotential surface.
- (C) Equipotential surfaces always enclose some volume.
- (D) Two equipotential surfaces do not intersect each other.

(iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : 1

- (A) 90° (B) 45°
- (C) 30° (D) 0°

OR

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : 1

- (A) remain stationary
- (B) move from A to B
- (C) move from B to A
- (D) move around Q in a circular path between the equipotential surfaces A and B

- (b) (i) Obtain an expression for the capacitance of a parallel plate capacitor.
- (ii) Two identical plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a distance d in air and a dielectric of dielectric constant K is filled between them. Write the expressions for :
- (1) the potential difference between the plates
 - (2) capacitance of the capacitor so formed
 - (3) energy stored in this capacitor